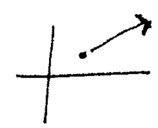
Vectors in $\mathbb{R}^2$

A <u>vector</u> in $\mathbb{R}^2$ has 2 components. It expresses magnitude + direction.
Think of it as having terminal and initial points

<u>Eg</u> The vector from $(1,2)$ to $(4,8)$ is $\langle 3, 6\rangle$



Note the angle bracket notation.
Also $3\vec{i} + 6\vec{j}$

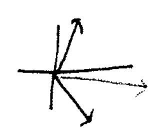
A regular old number is a <u>scalar</u>. We can add vectors and mult. by scalars.

<u>Eg</u> $\langle 3, 6\rangle + \langle -5, 7\rangle = \langle -2, 13\rangle$
$4\langle 3, 6\rangle = \langle 12, 24\rangle$

<u>Norm</u> $\|\langle 3,6\rangle\| = \sqrt{3^2 + 6^2} = \sqrt{45}$ Length of the vector
A <u>unit vector</u> has length 1.

$\left\langle \frac{3}{\sqrt{45}}, \frac{6}{\sqrt{45}} \right\rangle$ is a unit vector in same direction as $\langle 3, 6\rangle$

<u>Eg</u> Clueless Morgan + Polly Lobster pull Gonzo's Arms with force $\langle 3, 6\rangle$ N and $\langle 5, -9\rangle$ N, respectively.



The force on Gonzo is $\langle 3,6\rangle + \langle 5,-9\rangle = \langle 8,-3\rangle$ N

Vectors in $\mathbb{R}^3$

The algebra of vectors in $\mathbb{R}^3$ is the same as $\mathbb{R}^2$.
Each vector has 3 components $\langle 3, 6, -4 \rangle = 3\vec{i} + 6\vec{j} - 4\vec{k}$
These are the $x, y,$ and z - directions. We graph z as up + down.

$$\| \langle 3, 6, -4 \rangle \| = \sqrt{3^2 + 6^2 + (-4)^2} = \sqrt{61}$$

<u>Equation of a Sphere</u> $\quad (x-a)^2 + (y-b)^2 + (z-c)^2 = r^2$

<u>Eg</u> $\quad x^2 + 6x + y^2 + z^2 - 5z + 7 = 0$

$$(x+3)^2 + y^2 + (z - 5/2)^2 = -7 + 9 + \frac{25}{4} = \frac{33}{4}$$

Dot Product

Recall We can add vectors, but multiplying them doesn't work nicely.
Only as complex numbers (or quaternions, etc) does algebra work.

Eg $\langle 3,0\rangle$ "times" $\langle 0,4\rangle$ componentwise $= \langle 0,0\rangle$

$ab=0$ should imply a or $b=0$

We can multiply vectors and get a scalar, but if we try to get a vector, we must sacrifice some algebra.

Dot Product (Inner Product) $V \cdot W = V_1 W_1 + V_2 W_2 + V_3 W_3 = \|V\| \, \|W\| \cos\theta$

Eg $\langle 3,7,-2\rangle \cdot \langle 0,4,1\rangle = 0 + 28 - 2 = 26$

$\langle 3,0\rangle \cdot \langle 0,4\rangle = 0$ Notice $\rightarrow \cdot \uparrow$ they are orthogonal.

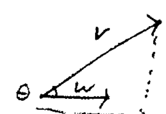
Properties
1. $V \cdot V = \|V\|^2$
2. $\vec{0} \cdot V = 0$
3. $u \cdot (v+w) = u \cdot v + u w$
4. $u \cdot v = 0 \iff u \perp v$

Eg Find angle btw. $\langle 3,7,-2\rangle$ and $\langle 0,4,1\rangle$.

$\|\langle 3,7,-2\rangle\| = \sqrt{9+49+4} = \sqrt{62}$ $\qquad \|\langle 0,4,1\rangle\| = \sqrt{17}$

$\theta = \cos^{-1}\left(\frac{26}{\sqrt{62}\sqrt{17}}\right)$

Defn Vector projection $\quad \mathrm{proj}_W V = \left(\frac{V \cdot W}{W \cdot W}\right) W = \|V\| \cos\theta \, \frac{W}{\|W\|}$



Scalar projection $\quad \mathrm{comp}_W V = \|\mathrm{proj}_W V\| = \frac{V \cdot W}{\|W\|}$ = how much V points in W direction

Eg $V = \langle 7,3,5\rangle \quad W = \langle 2,-2,-1\rangle \quad \mathrm{proj}_W V = \frac{-7}{9} W = \langle -\frac{14}{9}, \frac{14}{9}, \frac{7}{9}\rangle$

$\mathrm{comp}_W V = \frac{-7}{3}$

Cross Product

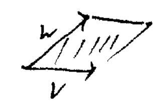
<u>Defn</u> $v \times w = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$ cross product

<u>Eg</u> $\langle 3,-1,3\rangle \times \langle 0,7,-4\rangle = \langle -17, 8, 14\rangle$

Notice the cross product produces a vector, but algebra suffers.

<u>Prop</u>

1. $\vec{v} \times \vec{v} = 0$
2. $\vec{v} \times w = -w \times v$
3. $(v \times w)\cdot v = (v \times w)\cdot w = 0$ <u>Right-hand Rule</u>
4. $u \times (v + w) = (u \times v) + (u \times w)$
5. $\|v \times w\| = \|v\| \, \|w\| \sin\theta$

<u>Eg</u> $\|v \times w\|$ = area of parallelogram 

$\frac{1}{2}\|v \times w\|$ = area of triangle 

<u>Scalar Triple Product</u> $(u \times v)\cdot w$ = Volume of parallelepiped 

$\begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$

<u>Torque</u> Suppose P is rotating around point O and a force F acts on P, tangentially. The torque is $\|OP \times F\|$

Work Old formula was Force times Distance

But often force is in one direction and motion in another.

Then Work = Force in the direction of motion times distance

$$(\text{comp}_w F)\ \|w\|$$

$$\text{Work} = \text{Force} \cdot \text{Distance}$$

Eg Clueless Morgan drags treasure chest along $\langle 10, 20, 0 \rangle$ (from origin to $(10, 20)$) by applying a force $\langle 4, 5, 3 \rangle N$.

$$\text{Work} = \langle 4, 5, 3 \rangle \cdot \langle 10, 20, 0 \rangle = 140 \text{ Joules}$$

Direction Cosines The direction angles are the 3 angles V makes with the coordinate axes. The direction cosines are their cosines.

Eg $V = \langle 3, 7, -2 \rangle$

$$\cos \alpha = \frac{3}{\|v\|} = \frac{3}{\sqrt{62}}$$

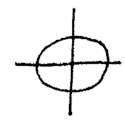
$$\cos \beta = \frac{7}{\|v\|} = \frac{7}{\sqrt{62}}$$

$$\cos \gamma = \frac{-2}{\|v\|} = \frac{-2}{\sqrt{62}}$$

Curves, Lines, Planes

<u>Parametric Equations</u> "Walking Ant" method of describing curves

<u>Eg</u> $X = \cos t \qquad y = \sin t \qquad 0 \le t \le 2\pi$



Eliminate parameter $\quad X^2 + y^2 = \cos^2 t + \sin^2 t = 1$

Recall Euler's Formula. This curve is also $e^{it}, \quad 0 \le t \le 2\pi$

<u>Lines</u> Find the line thru (X_0, Y_0, Z_0) in $\langle A, B, C \rangle$

$$(X_0, Y_0, Z_0) + t \langle A, B, C \rangle$$

$$X = X_0 + tA \qquad y = Y_0 + tB \qquad Z = Z_0 + tC$$

$$\frac{X - X_0}{A} = \frac{Y - Y_0}{B} = \frac{Z - Z_0}{C} \qquad (= t)$$ Symmetric Form

<u>Eg</u> Line thru $(1, -3, 2)$ in $\langle -2, 4, 1 \rangle$ direction

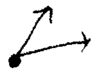
$$\frac{X - 1}{-2} = \frac{y + 3}{4} = \frac{Z - 2}{1} \qquad \boxed{\text{or}} \qquad \begin{aligned} X &= 1 - 2t \\ y &= -3 + 4t \\ Z &= 2 + t \end{aligned}$$

<u>Planes</u> Given a point (X_0, Y_0, Z_0) and a normal vector $\langle A, B, C \rangle$, the resulting plane is

<u>Pt-Normal</u> $\quad A(X - X_0) + B(y - Y_0) + C(Z - Z_0) = 0 \qquad$ (Dot Prod. $= 0$)

<u>Std</u> $\quad AX + By + CZ + D = 0$

Eg Find Eq. of plane thru $(1,2,3)$ $(-3,7,4)$ $(7,8,9)$


$(-3,7,4)$

Find two vectors in the plane $\langle 4,-5,-1\rangle$ $\langle 10,1,5\rangle$

Find a normal with cross product $\langle -24, 30, 54\rangle$

Plane: $-24(x-1)+30(y-2)+54(z-3)=0$

$\boxed{\text{or}}$

$-24x+30y+54z-148=0$